

Coincidence as a Bit of Information

Receiver-Defined Spatiotemporal Events, Phase-Wave Differentials, and the Sensory-to-Action Bridge

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Abstract

In April 2017, Micah Blumberg publicly proposed that a coincidence pattern could serve as a bit of information in the brain. His September 2018 clarification made the intended claim more precise than the later deposited paper: coincidence was the *basis* of an information unit, not a claim that one spike always equals exactly one Shannon bit; the relevant event could be subthreshold; and its information content could be greater or smaller than one bit. In 2022, the proposal was extended across synaptic, dendritic-branch, somatic-burst, and oscillating-network scales, then joined to the tonic-context / phasic-differential architecture of Self Aware Networks (SAN), Neural Array Projection Oscillation Tomography (NAPOT), and phase-wave differentials (PWDs). A 2024 book carried that joined mechanism into a sustained account of neural rendering.

This paper supplies the missing mathematical and experimental contract. It distinguishes (i) raw temporal co-occurrence, (ii) a receiver-defined detected event, and (iii) a causally used event. A receiver combines delayed, weighted, and phase-sensitive inputs against its current tonic context. Phase-compatible PWD inputs can add constructively; a thresholded receiver can convert that continuous resultant into a discrete event symbol. The symbol is binary only after the receiver, window, threshold, and alphabet have been declared. Its self-information is $-\log_2 p$, its entropy is at most one bit for a binary alphabet, and its task-relevant value is the conditional information it contributes beyond firing-rate, stimulus-history, and state baselines.

The resulting SAN hypothesis is a measurable chain: broad tonic coordination $\rightarrow$ phasic PWD events $\rightarrow$ phase-compatible summation and locking $\rightarrow$ increased effective receiver drive $\rightarrow$ motor-pool threshold crossing $\rightarrow$ neuromuscular transmission $\rightarrow$ muscle or vocal action $\rightarrow$ sensory return. Bat vocalization research provides a particularly useful architecture-level test case: pre-vocal frontal and striatal oscillations predict call class, frontal-to-auditory information flow predominates before vocalization, and echolocation produces a later bottom-up return. These findings do not themselves prove the SAN mechanism. They make it possible to test whether receiver-relative PWD variables add explanatory and predictive value beyond established rate, power, phase, and connectivity measures.

Keywords: coincidence detection; information theory; phase-wave differential; tonic and phasic activity; neural oscillations; dendritic computation; motor control; vocalization; Self Aware Networks; NAPOT

Scope and evidence boundary

Version note: This major revision expands the documentary genealogy, receiver-defined mathematics, reproducible examples, sensory-to-action mechanism, and empirical failure conditions of the original 2025 preprint.

1. The constructive claim

Imagine two sensory inputs arriving within a few milliseconds of one another. In one dendritic branch they may add strongly enough to change the cell's output. In another receiver, different conduction delays,

inhibition, or ongoing activity may keep the same pair below threshold. An analyst can call both pairs coincident, but only the first receiver turns that timing relation into a consequential event. The problem is to explain how a continuous pattern becomes a discrete, reusable symbol without pretending that every spike intrinsically carries one bit.

That problem requires three questions to be answered before the phrase *coincidence as a bit* earns a technical meaning.

1. What events count as coincident?
2. Which physical receiver makes that distinction?
3. How much information does the resulting event carry about a declared target?

In plain language, the proposed operation is this: a receiver weighs what arrived, when it arrived, and the state it arrived into; the receiver then either remains within its ongoing context or crosses into a new state that can affect downstream processing. The claim reconstructed from Blumberg’s dated corpus is a joined architecture:

A biological receiver can transform a continuous, weighted spatiotemporal input pattern into a discrete, context-relative event; that event can function as an information symbol inside a multiscale tonic / phasic rendering and action system.

Only after that operation is understood does the specialized vocabulary become useful. A continuous departure from the receiver’s ongoing context is called a *phase-wave differential* (PWD) when its phase and propagation relations are part of the specification. If the receiver partitions its possible states into a declared two-symbol alphabet, the resulting event can play the role of a *bit*. The word *bit* therefore names a role in an event alphabet, not an intrinsic quantity attached to every spike. The physical event may be a subthreshold dendritic response, a local spike, a burst, a synchronized population transition, or another declared receiver state. Information is then calculated from the event distribution and the variable it is supposed to encode.

This interpretation preserves the original idea while making it scientifically stricter. It also reveals why an earlier coincidence detector does not exhaust the comparison. Earlier work establishes important components. The Micah-side question continues through the coincidence-to-symbol transform, its multiscale placement, the tonic / PWD distinction, and the receiver-to-action loop.

2. Documentary genealogy

The source record supports a sequence of partial contributions rather than a single date attached to the fully mature theory.

Date	Recovered contribution	Boundary
2011	Prediction-pattern self and whole-network observer; internally dated stereographic-camera note describes distributed “pixels,” learned statistical links, inference, prediction, and one continuously updated multisensory picture	Functional rendering ancestor; internal 2011 date is preserved separately from June 2022 public Git fixation; not yet the coincidence-bit formulation
2012	EEG / event synchronization, multimodal co-activity, and out-of-sync non-linking	Temporal-alignment ancestor; not PWD
2017-04-21	Public proposal that a brain bit can be a “coincidence pattern”	Explicit coincidence-as-information-unit node
2018-05-30	Peter Tse named as a source for neurons as coincidence detectors	Influence is explicit, not hidden

Date	Recovered contribution	Boundary
2018-09-24	Coincidence is the basis of a bit; not one spike = one bit; subthreshold events may count	Establishes the intended non-naive interpretation
2021-02-09	Multivesicular-release discussion challenges a one-vesicle / one-symbol caricature	Graded synaptic bridge; not the full PWD claim
2022-06-08	Coincidence hierarchy spans synaptic firing, dendritic branches, and somatic bursts	Explicit multiscale event hierarchy in immutable Git
2022-09-18	Tonic same-frequency patterns and high-phasic burst printing are joined to phase-change information	Explicit tonic-context / phasic-differential architecture
2024-10-30	Book-length integration of high-phasic PWDs with slower tonic context, dendrites, coincidence, and rendering	Mature published synthesis
2025-06	Deposited historical paper and later SAN / SIT synthesis	Public consolidation; same-author development is not independent validation
2026-07	Receiver-relative equations, conditional-information test, causal-use criterion, and motor bridge	Version 2 formalization

References [1-7] identify the principal public first-party routes. The accompanying reproducibility supplement supplies the public-path-safe custody table, immutable-byte hashes, date types, and genealogy ledger. The chronology makes three constraints explicit. First, mature names such as NAPOT and PWD are not backdated into earlier wording. Second, an internal date and a later public Git fixation are separate provenance facts; the 2011 stereographic-camera note is therefore evidence of conceptual genealogy, while June 2022 is its recovered immutable public boundary. Third, the absence of the complete 2026 formalism in 2017 does not erase the 2017 information-unit proposition or the later dated additions. Each atom is dated to the earliest source that actually contains it, with its date type and custody boundary stated.

3. Earlier scientific components and the remaining delta

The central claim of this paper is not that timing matters, that neurons can detect coincident input, or that oscillatory phase can affect communication. Each has a substantial earlier scientific history.

Softky and Koch analyzed high-variance cortical spiking in terms of coincidence detection in 1993 [9]. König, Engel, and Singer argued in 1996 that short integration intervals make cortical neurons behave as coincidence detectors that preferentially relay synchronized input [10]. Spike-timing-dependent plasticity experiments linked relative timing to synaptic change [11]. Dendritic studies established nonlinear interactions between inputs arriving at different cellular compartments [12,13]. Communication-through-coherence and related models examined how phase relations regulate effective interaction between populations [14]. Phase-of-firing studies showed that spike phase can provide information beyond spike count [15,16].

These are named earlier components, not an author-level ownership boundary over every later composite containing coincidence, timing, phase, or dendrites. The remaining Micah-side delta is the dated combination of:

- a coincidence pattern explicitly interpreted as the basis of an information unit;
- rejection of an obligatory one-spike / one-bit equivalence;

- inclusion of subthreshold and multiscale receiver events;
- placement of those events inside a persistent tonic context and phasic PWD update process;
- propagation from distributed sensory evidence through choice and motor realization;
- a proposed common vocabulary for biological and machine receivers.

No claim is made here that the exact phrase had never appeared elsewhere. The documentary result is that Blumberg publicly used and clarified this formulation on the cited dates, then developed the joined architecture through later dated sources.

4. Three event levels

4.1 Raw co-occurrence

Let $N_i(t)$ be the counting process for input channel i , with event train $n_i(t) = \sum_j \delta(t - t_{ij})$. For a window of duration Δ and a minimum count k , a raw co-occurrence indicator can be defined as

$$Q_t(\Delta, k) = \mathbf{1} \left\{ \sum_{i=1}^m \mathbf{1}[N_i(t) - N_i(t - \Delta) > 0] \geq k \right\}. \quad (1)$$

This is a statistic chosen by an analyst. It does not yet identify a biological receiver, its delays, its synaptic signs, or its current excitability.

4.2 Receiver-defined detected event

For receiver R , let w_{Ri} be an excitatory or inhibitory weight, τ_{Ri} a conduction delay, K_{Ri} an integration kernel, and $\mathcal{H}_R(t)$ the receiver's observable history and state. Using the event train defined above, a simple receiver drive is

$$u_R(t) = \sum_{i=1}^m w_{Ri} \int_0^{\Delta_R} K_{Ri}(s) n_i(t - s - \tau_{Ri}) ds - b_R(t | \mathcal{H}_R). \quad (2)$$

Here b_R is a state-dependent baseline or predicted tonic input. The receiver-defined event is

$$C_R(t) = \mathbf{1}\{u_R(t) \geq \theta_R(t | \mathcal{H}_R)\}. \quad (3)$$

Equations (2)-(3) make coincidence relational. The same incoming spikes can count as an event for one dendritic branch, neuron, or population and not for another because the receivers differ in connectivity, delay, inhibition, threshold, and state.

4.3 Causally used event

A detected event is not necessarily used by the circuit. Let $Y_R(t + \ell)$ be a downstream response at lag ℓ . Observational dependence is insufficient for causal use. A receiver event is causally used only if a targeted intervention that changes the event while preserving relevant controls changes the downstream response:

$$\mathcal{U}_R = 1 \quad \text{only if} \quad P(Y_R | do(C_R = 1), B) \neq P(Y_R | do(C_R = 0), B), \quad (4)$$

where B is the preregistered control set. This distinction prevents a correlated coincidence from being mislabeled a functional neural symbol.

5. Tonic context and the high-phasic PWD

5.1 Context is not a frequency label

In SAN, *tonic* names the comparatively persistent context against which a change is registered. *Phasic PWD* names the comparatively event-like differential. This is not a universal equation of “tonic” with one fixed low-frequency band or “PWD” with one fixed high-frequency band. Temporal frequency, amplitude, spatial extent, propagation speed, and persistence must be measured separately.

Let $x_R(t)$ be a receiver-scale signal and K_T a nonnegative, normalized longer-context kernel, so that $\int_0^{W_T} K_T(s) ds = 1$. The tonic estimate is

$$T_R(t) = \int_0^{W_T} K_T(s) x_R(t-s) ds, \quad W_T > \Delta_R. \quad (5)$$

The receiver-relative differential is

$$d_R(t) = x_R(t) - T_R(t). \quad (6)$$

The longer context window and shorter event window implement the two-timescale distinction: a broad, persistent coordination state supplies the canvas; a temporally compressed departure supplies candidate new information. A sharp-wave-ripple-associated volley, for example, can rapidly recruit distributed targets and replay compressed sequences within slower state-dependent coordination [17]. The point is not that every high-frequency field oscillation travels farther than every slow rhythm. It is that a brief coordinated event and a persistent coordinating context have different temporal jobs.

5.2 Phase-compatible PWD summation

Represent the (i)th incoming PWD at receiver (R) by a complex phasor after its weight, delay, and receiver phase preference have been applied:

$$z_{Ri}(t) = w_{Ri} a_i(t - \tau_{Ri}) e^{i[\phi_i(t - \tau_{Ri}) - \psi_{Ri}(t)]}. \quad (7)$$

The coherence-weighted resultant is

$$Z_R(t) = \sum_{i=1}^m z_{Ri}(t), \quad A_R(t) = |Z_R(t)|. \quad (8)$$

An alignment index is

$$\rho_R(t) = \frac{|\sum_i z_{Ri}(t)|}{\sum_i |z_{Ri}(t)|}, \quad 0 \leq \rho_R(t) \leq 1. \quad (9)$$

Equation (9) applies when $\sum_i |z_{Ri}(t)| > 0$; when no modeled input amplitude is available, this paper sets $\rho_R(t) = 0$ by convention.

When equal inputs arrive with compatible phase, their resultant can approach the sum of their amplitudes. When they occupy a balanced splay arrangement, their signed resultant can be small even though every input

is active. Phase locking does not mean the oscillators exert no influence on one another; stable relative phase is ordinarily maintained through coupling and correction.

The sum in Equations (7)-(9) requires a declared narrowband component or another explicit mapping into a shared receiver-relative phase coordinate. Raw phases from unrelated frequencies cannot be added as though they were already commensurate.

5.3 Electrical valence as effective drive

This paper uses *electrical valence* operationally, not metaphorically. It is the signed, coherence-weighted drive available to influence a declared receiver after excitation, inhibition, delay, and current state are included:

$$V_R(t) = \int_0^{\Delta_R} K_D(s) \operatorname{Re}[Z_R(t-s)] ds - \lambda_R I_R(t), \quad (10)$$

where K_D is a nonnegative receiver-drive kernel supported on $[0, \Delta_R]$ and normalized so that $\int_0^{\Delta_R} K_D(s) ds = 1$. The integral therefore has the units of the calibrated Z_R signal. The product $\lambda_R I_R(t)$ must be expressed in those same units. It is an additional inhibitory contribution and must contain only inhibition not already represented by signed weights in Z_R ; if all inhibition is already included there, set $\lambda_R I_R(t) = 0$ to avoid double counting. A candidate event occurs when that drive crosses a receiver threshold:

$$C_R(t) = \mathbf{1}\{V_R(t) \geq \theta_R(t)\}. \quad (11)$$

This is the formal version of the proposed chain: concurrent, phase-compatible PWD activity can increase effective drive and select which receiver crosses threshold. The term does not identify electrical valence with psychological valence, bulk charge, LFP power, or energy. Those are different variables unless an experiment explicitly connects them.

The symbol V_R denotes this valence index, not physical volts. Its units inherit from the calibrated receiver signal and kernel; comparisons across receivers require a declared normalization or biophysical conversion.

6. How many bits?

The following distinctions use Shannon's mathematical framework while keeping event definition and biological implementation separate [8].

6.1 Event self-information

For a declared event outcome c and baseline value $B = b$, its conditional surprisal is

$$\iota(c \mid B = b) = -\log_2 P(C_R = c \mid B = b). \quad (12)$$

A rare event can have more than one bit of self-information. This does not mean the receiver has a binary alphabet with entropy greater than one bit per sample; self-information and entropy are different quantities.

6.2 Binary entropy

If $C_R \in \{0, 1\}$ with $P(C_R = 1) = p$, then

$$H(C_R) = -p \log_2 p - (1 - p) \log_2 (1 - p) \leq 1. \quad (13)$$

Using the standard convention $0 \log_2 0 = 0$, equality holds only at $p = 1/2$. Thus “coincidence as a bit” can defensibly mean that a receiver partitions its input into a two-symbol event alphabet. It cannot mean that every event intrinsically carries exactly one bit of entropy.

6.3 Information about a target

For stimulus, state, decision, or action variable S , the event’s mutual information is

$$I(S; C_R) = \sum_{s,c} P(s, c) \log_2 \frac{P(s, c)}{P(s)P(c)}. \quad (14)$$

The stronger test is incremental information beyond baseline B , where B can contain firing rate, recent spike history, oscillatory power, movement, stimulus history, and receiver state:

$$\Delta I_R = I(S; C_R \mid B). \quad (15)$$

This is the quantity the biological hypothesis must earn. If ΔI_r is zero after honest controls, the proposed coincidence variable adds no information for that target. If it is positive on held-out data, it is a candidate code but still not a causal mechanism.

For a downstream outcome Y , a directed observational statistic can be written

$$I(C_R(t); Y(t + \ell) \mid Y_{<t}, B), \quad (16)$$

but it should not be called causal without the intervention in Equation (4) or a justified identification design.

7. Deterministic examples

The accompanying implementation evaluates three deliberately small examples. They demonstrate the definitions; they are not fitted to neural data.

7.1 Independent-Poisson coincidence

For four independent channels with equal rate r , a 5 ms window, and a two-channel threshold, the probability that one channel is active is

$$q = 1 - e^{-r\Delta}, \quad (17)$$

and the receiver-event probability under this null is

$$P(C_R = 1) = \sum_{j=2}^4 \binom{4}{j} q^j (1-q)^{4-j}. \quad (18)$$

The exact implementation gives:

Channel rate	Receiver-event probability	Event surprisal	Binary entropy
5 Hz	0.003538	8.1427 bits	0.03391 bits
20 Hz	0.047687	4.3903 bits	0.27649 bits
50 Hz	0.214172	2.2232 bits	0.74938 bits

The same event rule produces different information values as the background rate changes. A rate-preserving jitter or conditional point-process null is therefore essential in real data.

7.2 Phase-compatible summation

Four unit-amplitude PWD inputs at the receiver’s preferred phase produce $A_R = 4$, $\rho_R = 1$, and signed drive $+4$. Four equally aligned inputs at phase π relative to that receiver also have magnitude 4, but their signed drive is -4 , so they do not cross a positive threshold. Four inputs at phases 0 , $\pi/2$, π , and $3\pi/2$ produce a resultant numerically indistinguishable from zero. Under a declared receiver threshold of 3.5, only the receiver-preferred aligned case crosses the gate. This separates phase alignment from receiver-specific signed drive; whether a real receiver implements the assumed phasor readout is an empirical question.

7.3 Conditional information

In the declared binary example, the baseline already predicts the target, but the receiver event retains 0.1370 bits of conditional mutual information. The example shows why an event must be tested against the strongest reasonable baseline rather than against chance alone.

All eleven deterministic checks pass in normal and optimized execution. The code, parameters, and machine-readable results accompany this draft.

8. From sensory evidence to choice and action

8.1 The proposed chain

Figure 1. The proposed tonic-context $\rightarrow$ phasic-PWD $\rightarrow$ receiver-drive $\rightarrow$ motor-realization chain. The diagram separates neural selection and timing from the ATP-powered energy of contraction.

The joined SAN / PWD mechanism is

$$\begin{aligned} \text{tonic context} &\rightarrow \text{phasic PWD} \rightarrow \text{phase-compatible receiver summation} \rightarrow \text{effective drive} \\ &\rightarrow \text{selected motor pool} \rightarrow \text{action} \rightarrow \text{sensory return.} \end{aligned} \quad (19)$$

Coincident arrival is physiologically relevant because synaptic inputs sum over finite membrane and dendritic windows. Common synaptic input to motor-neuron groups can organize recruitment, while firing rate and motor-unit recruitment regulate force [18,19]. Once a motor-neuron action potential reaches the neuromuscular junction, calcium-dependent acetylcholine release produces an end-plate potential; the

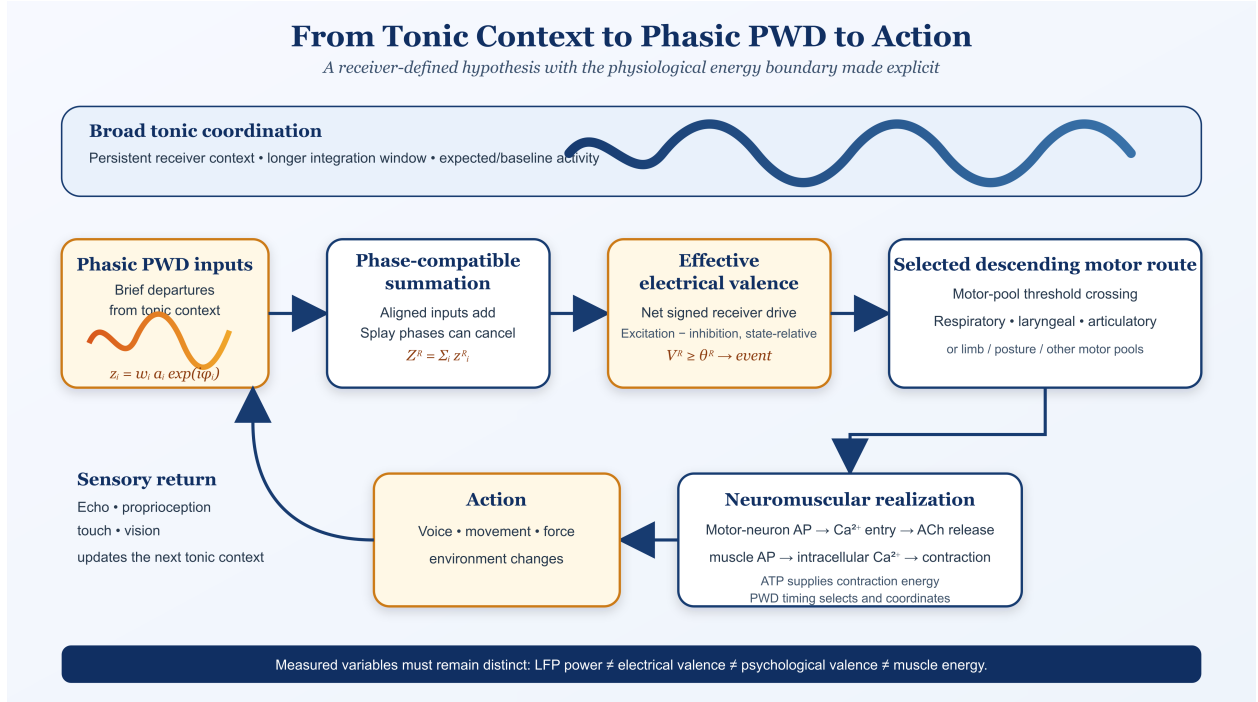


Figure 1: Figure 1. The proposed tonic-context → phasic-PWD → receiver-drive → motor-realization chain. The diagram separates neural selection and timing from the ATP-powered energy of contraction.

muscle action potential then initiates excitation-contraction coupling and intracellular calcium release [20,21].

Vocal realization is a coordinated special case rather than a single “vocal-cord threshold.” Descending pathways must organize respiratory pressure, laryngeal motor control, and - where relevant - articulatory movements. Their convergent neural inputs help determine when the corresponding motor neurons fire; the neuromuscular junction converts those motor-neuron impulses into muscle-fiber excitation; calcium release and ATP-dependent cross-bridge cycling produce contraction. Concurrent or common drive can coordinate and grade the recruited muscle groups, but it is not necessary to claim that every contributing motor neuron must fire in perfect synchrony.

The PWD does not provide the metabolic energy of contraction. The hypothesis is that its relative timing and phase structure help determine which distributed pathway gains effective control over the final common motor route.

8.2 Bat vocalization as an architecture-level test case

Weineck and colleagues recorded frontal-cortex and dorsal-striatal activity in vocalizing bats [22]. Pre-vocal beta- and high-gamma-band activity changed before call onset and helped predict whether the upcoming sound would be echolocation or communication. The authors explicitly described the result as correlational rather than proof that the oscillations caused vocal output.

García-Rosales and colleagues then measured the fronto-auditory loop [23]. Transfer-entropy analyses found predominant frontal-to-auditory flow during spontaneous and pre-vocal periods. After echolocation emission, the preferred direction shifted toward auditory-to-frontal flow, consistent with self-generated acoustic feedback. Frontal stimulation also changed auditory-cortex responses.

Together, these papers instantiate an important narrative architecture:

pre-action\frontal/striatal\oscillatory\state\→\impending\vocal\class\→\vocal\action\→\echo-driven\sensory\return

The coincidence-to-action components are therefore not a SAN-only phenomenon. Across the two studies, the bat literature already measures the main observables needed by the proposed mechanism: band-specific power, spike-LFP locking, interregional coupling, directional information flow, prediction of the selected vocal route, emitted action, and sensory return. SAN’s claimed contribution is the joined interpretation of those observables as one sensory → choice → action → feedback operator involving PWD and accumulated effective electrical valence.

The studies do not directly measure SAN’s V_R , and beta power should not simply be renamed electrical valence. They provide the right experimental setting for asking whether the joined coherence-weighted, receiver-relative PWD variable explains pre-vocal selection better than power, firing rate, phase, and connectivity modeled separately.

The strongest test would use the available pre-vocal recordings or a new preregistered dataset to compare:

- band power and spike rate;
- pairwise phase locking;
- receiver-specific delayed phasor resultants;
- conditional information about call class;
- perturbation of the predicted high-valence receiver state;
- matched rescue by restoring the predicted phase relation.

9. Experimental program

9.1 Required data

A credible test needs simultaneous sender and receiver recordings, sufficient temporal precision, identified anatomical direction where claimed, behavior or stimulus labels, motor output, and a sensory-return channel. Candidate preparations include apical / basal dendritic coincidence, sensorimotor cortex with EMG, bat vocalization, or controlled speech-motor tasks.

9.2 Frozen event specification

Before opening the final test set, preregister:

1. receiver identity and scale;
2. input channels and anatomical or statistical delays;
3. tonic-context window W_T ;
4. phasic event window Δ_r ;
5. weights, inhibitory signs, and phase preference;
6. event threshold and alphabet;
7. baseline covariates;
8. primary target and lag;
9. null and surrogate families;
10. perturbation and rescue endpoints.

9.3 Model comparison

At minimum compare:

- rate / history baseline;
- power-only baseline;
- phase-only baseline;
- connectivity-plus-power model;
- coincidence-count model;
- receiver-relative PWD model;
- combined model.

Evaluate held-out log loss, calibration, conditional mutual information, and effect-size uncertainty. Nested cross-validation must contain all threshold, delay, frequency-band, and receiver selection. Multiple receivers and lags require multiplicity control.

9.4 Causal discriminator

The distinctive test is not merely decoding. Intervene on the predicted phase relationship while holding mean input count and power as closely matched as feasible. The SAN / PWD account predicts:

- phase-compatible stimulation should increase the declared receiver resultant and event probability;
- a balanced splay manipulation should reduce the resultant without necessarily reducing total input amplitude;
- changing the receiver's tonic state should change which identical phasic input counts as a large differential;
- disrupting the predicted receiver event should alter the downstream choice or action;
- restoring the phase relation should rescue the effect.

9.5 Failure conditions

The proposed operator fails in a tested preparation if, after adequate power and validated measurement:

- the receiver-relative PWD variable adds no out-of-sample information beyond rate, power, history, and connectivity;
- splay and aligned inputs with matched amplitude produce no predicted receiver difference;
- tonic-context manipulation does not alter differential sensitivity;
- selective disruption of the event leaves the predicted downstream action unchanged;
- a simpler model explains the same result with equal or better calibration and complexity penalty.

Failure in one preparation limits that instantiation; it does not prove that every possible receiver lacks such an operation. Conversely, one positive preparation does not establish a universal brain code.

10. Relationship to SAN, NAPOT, and SIT

Within SAN, receiver-defined coincidences are local decisions inside a distributed observer. Within NAPOT, they can contribute to how partial sensory, memory, and recurrent signals are integrated and projected across neural arrays. Within PWD, the informational object is the receiver-relative departure from an expected tonic context, including its phase, amplitude, duration, frequency, and propagation consequences.

This paper therefore sharpens the relationship among three claims:

1. **Coincidence event:** a receiver-defined partition of spatiotemporal input.

2. **PWD:** a continuous, context-relative change that can create or modify such an event.
3. **Information bit:** a discrete symbol assigned after the receiver and alphabet are specified.

SIT is a later same-author extension into physics and thermodynamics. It can supply a broader research program, but it cannot independently validate the 2017 neural proposal. No inference from the present equations establishes gravity, cosmology, quantum coherence, or a universal information ontology.

11. Claim ledger

Claim	Current status
Blumberg publicly used coincidence-pattern / bit wording in April 2017	Documented, bounded
Blumberg explicitly rejected one-spike = exactly one-bit in September 2018	Documented, bounded
The proposal later expanded across synaptic, dendritic, burst, tonic, and PWD scales	Documented as same-author genealogy
A receiver-defined coincidence creates a discrete event variable	Defined mathematically
Every coincidence intrinsically carries exactly one bit	Rejected by the mathematics and by the 2018 clarification
Phase-compatible phasors can add while splay phasors cancel	Mathematical result for the declared model
Neural receivers implement the exact phasor model in Equations (7)-(11)	Open experimental hypothesis
Pre-vocal bat oscillations prove SAN electrical valence	Not established; they motivate a discriminating test
Receiver-relative PWD predicts action beyond rate, power, history, and connectivity	Open empirical question
Coincidence / PWD timing directly supplies muscle energy	Not claimed

12. Conclusion

The recovered record supports a stronger and more precise result than either a slogan or a retreat. In 2017, Micah Blumberg publicly proposed coincidence pattern as the basis of a neural information unit. In 2018, he clarified that the proposal was not one spike equals exactly one bit and that subthreshold events could qualify. In 2022, he placed coincidence at multiple cellular scales and joined it to tonic oscillatory context and high-phasic differential updates. In 2024, that tonic-canvas / PWD architecture became part of a book-length SAN account.

The formal contribution of this revision is to turn that genealogy into a testable receiver contract. Coincidence becomes a context-sensitive transformation from weighted spatiotemporal input to a discrete event; phase-compatible PWDs can add into an effective signed drive; information is measured conditionally against strong baselines; and causal use requires perturbation. The sensory-to-choice-to-action loop supplies the decisive empirical arena. Bat vocalization and motor-pool physiology already provide component measurements. The next step is to test whether the joined receiver-relative operator explains and controls behavior better than existing rate, power, phase, and connectivity descriptions.

Data, code, rights, and release status

The deterministic examples use no human or animal subject data. Private custody material, including local paths, remains outside the public release. The Version 2 reproducibility supplement contains public-path-safe source records, model parameters, code, results, tests, and figure sources. Third-party papers are cited rather than redistributed. Independent information-theory, computational-neuroscience, archival-source, and rights review remains invited.

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